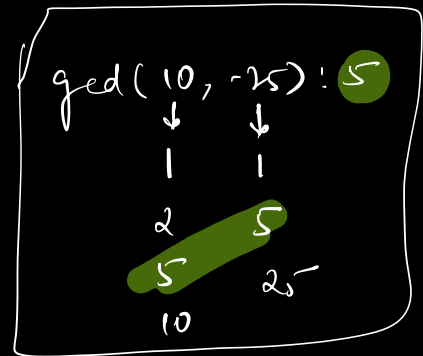
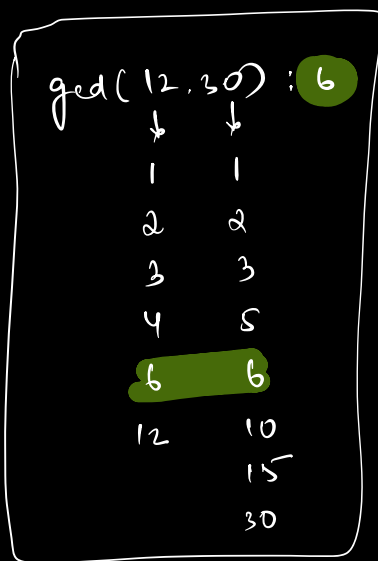
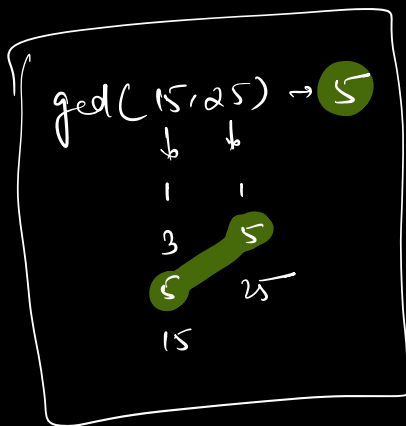


gcd $\rightarrow$ Greatest Common Divisor

$\downarrow$
HCF $\rightarrow$ Highest Common Factor

$\text{gcd}(a, b) \rightarrow$ greatest factor which divides both a & b .

$\text{gcd}(a, b) = x$ $a \div x = 0$ & $b \div x = 0$, x is greatest such no.



$$\begin{array}{r} 25 \\ 5 \overline{) 25} \\ \underline{25} \\ 0 \end{array} \Rightarrow 0 \text{ remainder}$$

$$\begin{array}{r} 25 \\ 5 \overline{) 25} \\ \underline{25} \\ 0 \end{array}$$

$$\begin{array}{r} 25 \\ 5 \overline{) 25} \\ \underline{25} \\ 0 \end{array}$$

$$-25 \div -5 = 0$$

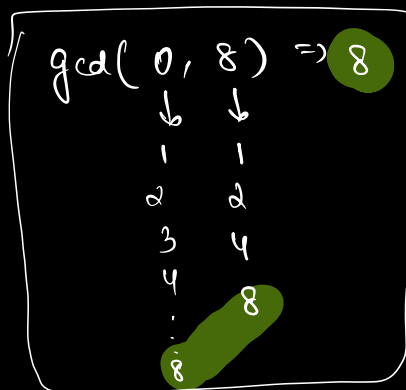
$$-25 \div -5 = 0$$

$$-25 \div -1 = 0$$

$$-25 \div 1 = 0$$

$$-25 \div 5 = 0$$

$$-25 \div 25 = 0$$



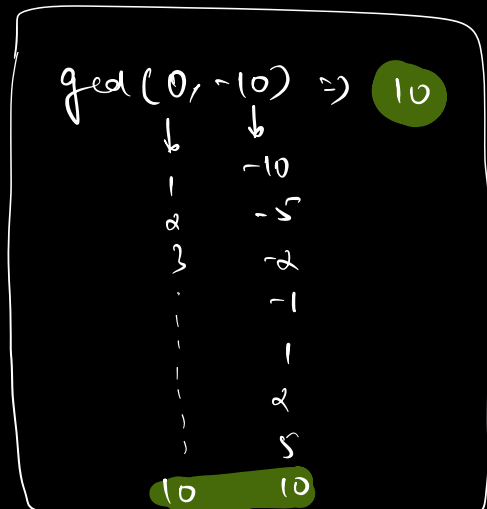
$$0 \div 1 = 0$$

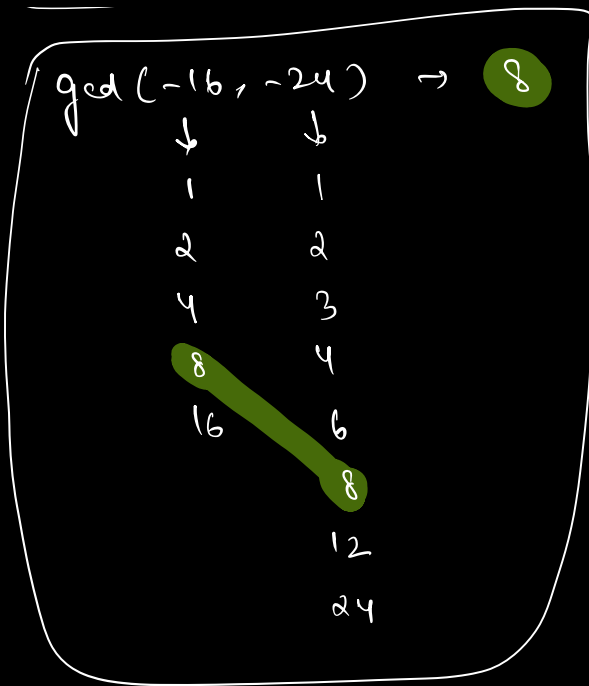
$$0 \div 2 = 0$$

$$0 \div 3 = 0$$

$$0 \div 4 = 0$$

$$0 \div 8 = 0$$





$$-16 \div -2 = 0 \quad \checkmark$$

$$\begin{array}{r} -2 \overline{) -16} \quad 8 \\ \underline{-16} \\ 0 \end{array}$$

$$-16 \div 2 = 0 \quad \checkmark$$

$$\begin{array}{r} 2 \overline{) -16} \quad -8 \\ \underline{-16} \\ 0 \end{array}$$

x is -ve

$$\left[\begin{array}{l} x \div -a \geq 0 \\ x \div a \geq 0 \end{array} \right]$$

Properties of gcd: $|| \rightarrow$ abs value / modulus (+ve)

i) $\text{gcd}(a, b) \Rightarrow \text{gcd}(|a|, |b|)$

ii) $\text{gcd}(a, b) \Rightarrow \text{gcd}(b, a)$

iii) $\text{gcd}(0, x) \Rightarrow |x|$

if $x=0$, $\text{gcd}(0, 0) \Rightarrow$ infinite / undefined.

iv) $\text{gcd}(a, b, c) \Rightarrow \text{gcd}(\text{gcd}(b, c), a)$

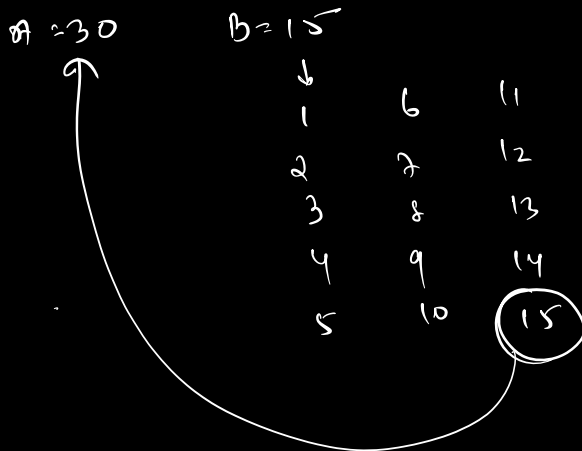
$\downarrow$
Associative

$$\text{gcd}(\text{gcd}(a, c), b)$$

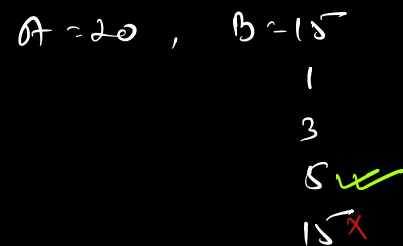
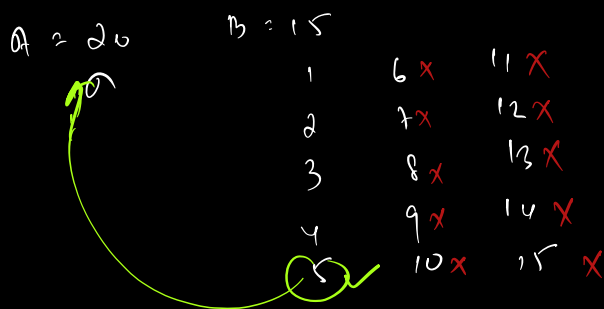
$$\text{gcd}(\text{gcd}(b, a), c)$$

Q1. Given 2 nos. A & B. Find $\text{gcd}()$?

Soln 1



$|x| \Rightarrow +ve$
 $\downarrow$
abs.
 $\% \Rightarrow$ remainder
 $\downarrow$
mod



$\text{gcd}(A, B) \{$

A = |A|

B = |B|

for (i = min(A, B); i >= 1; i--) {

if (A % i == 0 && B % i == 0) {

return i

}

}

}

TC $\Rightarrow O(\min(A, B))$

// check only for
the factors of

$\min(A, B)$

↓

TC $\Rightarrow O(\sqrt{\min(A, B)})$

• Euclidean Algo:-

if $a, b \Rightarrow (a > b)$

$$\boxed{\gcd(a, b) = \gcd(a-b, b)}$$

given

$\rightarrow \gcd(a, b) = x \rightarrow a \% x = 0 \text{ \& \& } b \% x = 0$

$\gcd(a-b, b) = x \rightarrow (a-b) \% x = 0 \text{ \& \& } b \% x = 0$

$$\begin{aligned} & \downarrow \\ & (a \% x - b \% x + x) \% x \\ & \quad \downarrow \quad \downarrow \\ & \quad 0 \quad 0 \\ & \Rightarrow (x) \% x \\ & \Rightarrow 0 \quad \text{LHS} = \text{RHS proved} \end{aligned}$$

$$\boxed{\begin{aligned} & \text{if } (a > b) \\ & \quad \gcd(a, b) = \gcd(a-b, b) \end{aligned}}$$

$a > 0, b > 0$

$\gcd(a, b) \{$

 if $(a < b) \{$

 swap(a, b)

 }

 if $(b == 0)$ return a

 return $\gcd(a-b, b)$

}

$\gcd(8, 6) \{$

 return $\gcd(2, 6)$

}

$\gcd(2, 6) \{$

 swap(a, b) $\Rightarrow a = 6, b = 2$

 return $\gcd(4, 2)$

}

$\gcd(4, 2) \{$

 return $\gcd(2, 2)$

}

$\gcd(2, 2) \{$

 return $\gcd(0, 2)$

}

$$T.C \Rightarrow O(\max(A, B))$$

$$\gcd(16, 1)$$

↓

$$\gcd(15, 1)$$

↓

$$\gcd(14, 1)$$

↓

$$\gcd(13, 1)$$

↑

$$(0, 1)$$

$$\gcd(\overset{a}{0}, \overset{b}{2}) \{$$

$$\text{swap} \Rightarrow a=2, b=0$$

$$\text{return } \gcd(2, 0)$$

}

$$33 - 5 = 28$$

$$28 - 5 = 23$$

$$23 - 5 = 18$$

$$18 - 5 = 13$$

$$13 - 5 = 8$$

$$8 - 5 = 3$$

$$33 \% 5 = 3$$

$$33 \% 5 = 33 - 6 \times 5$$

$$\begin{pmatrix} a \\ 23 \end{pmatrix}, \begin{pmatrix} b \\ 5 \end{pmatrix} \rightarrow \begin{pmatrix} a \\ 18 \end{pmatrix}, \begin{pmatrix} b \\ 5 \end{pmatrix} \rightarrow \begin{pmatrix} a \\ 13 \end{pmatrix}, \begin{pmatrix} b \\ 5 \end{pmatrix} \rightarrow \begin{pmatrix} a \\ 8 \end{pmatrix}, \begin{pmatrix} b \\ 5 \end{pmatrix} \rightarrow \begin{pmatrix} a \\ 3 \end{pmatrix}, \begin{pmatrix} b \\ 5 \end{pmatrix}$$

$$23 \% 5$$

$$\gcd(a, b) \rightarrow \gcd(a-b, b)$$

$$\gcd(a-2b, b)$$

$$\gcd(a-3b, b)$$

⋮

$$\text{gcd}(a, b) \Rightarrow \text{gcd}(a \% b, b)$$

$$\underline{\underline{a \% b < b}}$$

if ($a > b$)

$$\text{gcd}(a, b) \Rightarrow \text{gcd}(b, a \% b)$$

```

gcd(a, b) {
    if (b == 0)
        return a
    return gcd(b, a % b)
}
    
```

$$\begin{array}{c}
 \overset{a}{\text{gcd}}(\overset{b}{15}, 6) \\
 \swarrow \searrow \text{15 \% 6} \\
 \text{gcd}(6, 3) \\
 \swarrow \searrow \text{6 \% 3} \\
 \text{gcd}(3, 0) \\
 \downarrow \\
 \textcircled{3}
 \end{array}$$

$$\begin{array}{c}
 \overset{a}{\text{gcd}}(\overset{b}{6}, \overset{a}{21}) \\
 \swarrow \searrow \text{6 \% 21} \\
 \text{gcd}(21, 6) \\
 \swarrow \searrow \text{21 \% 6} \\
 \text{gcd}(6, 3) \\
 \swarrow \searrow \text{6 \% 3} \\
 \text{gcd}(3, 0) \\
 \downarrow \\
 \textcircled{3}
 \end{array}$$

$$\text{gcd}(a, b) = \text{gcd}(b, a \% b)$$

$$\begin{array}{c}
 \textcircled{a \% b} < b \\
 [0 \text{ to } b-1]
 \end{array}$$

$$b \leq a_2$$

$$\begin{array}{l}
 a \% b \Rightarrow < b \\
 \left(\begin{array}{l} b \leq a_2 \\ a \% b < a_2 \end{array} \right)
 \end{array}$$

$$O(\log(\max(a, b)))$$

$$b > a_2$$

$$\underline{\underline{a \% b = a - nb}}$$

$$n = 1$$

$$\underline{\underline{a - b}} \Rightarrow a - (> a_2) \Rightarrow < a_2$$

$$\underline{\underline{a - b < a_2}}$$

$$n = 2 \quad \underline{\underline{a - 2b}} \Rightarrow \underline{\underline{-ve}}$$

$$\boxed{a \% b = a - b \times a_2}$$

$$a \% b < a_2$$

$$O(\log(\max(a, b)))$$

$$TC \Rightarrow O(\log(\max(a, b)))$$

Answer

Q2. Given N array elements, find gcd of factorials of all given element.

ex $\Rightarrow$ $\begin{bmatrix} 4 & 3 & 8 & 6 \end{bmatrix}$
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
gcd $\begin{bmatrix} 24 & 6 & 40320 & 720 \end{bmatrix} \rightarrow$

$4! \rightarrow 1 \times 2 \times 3 \times 4$

$3! \rightarrow 1 \times 2 \times 3$

$8! \rightarrow 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$

$6! \rightarrow 1 \times 2 \times 3 \times 4 \times 5 \times 6$

$$\text{GCD} \Rightarrow 1 \times 2 \times 3 = 6 \Rightarrow 3!$$

$$O(n) \Rightarrow \text{factor}(\text{min element of arr}) \rightarrow O(\sqrt{n})$$

Subseq:

A subseq. can be generated from an array by

deleting 0 or more elements from array

$$\text{arr}[n] \Rightarrow \{ \overset{0}{3} \overset{1}{2} \overset{2}{1} \overset{3}{6} \overset{4}{4} \overset{5}{8} \overset{6}{10} \overset{7}{9} \}$$

$$\hookrightarrow \{ 3 \ 2 \ 6 \ 10 \ 9 \} \rightarrow \checkmark$$

$$\{ 2 \ 6 \ 10 \ 9 \} \rightarrow \checkmark$$

Subseq $\rightarrow$ may or may not be contiguous.

order has to be maintained

Q2. Given an array, return true if there exists any subseq whose $\text{GCD} = 1$

$$\text{ex} \Rightarrow 4 \ 6 \ 3 \ 8 \Rightarrow \{ 4 \ 3 \ 8 \} \Rightarrow \text{gcd}(4, 3, 8) = 1.$$

$\checkmark$

$$16 \ 10 \ 6 \ 15 \ 27 \Rightarrow \{10, 27\} \Rightarrow \gcd(10, 27) \\ = 1$$

$$6 \ 12 \ 3 \ 18 \Rightarrow \text{XX}$$

Ans

Assume $\Rightarrow$ subseq $\gcd[B, C, E] = 1$

$$\text{arr} \Rightarrow [A \ \underline{B} \ \underline{C} \ D \ \underline{E} \ F]$$

$$\gcd(A, B, C, D, E, F)$$

$$= \gcd(\gcd(B, C, E), \gcd(A, D, F))$$

$$= \gcd(1, \gcd(A, D, F))$$

$$= 1$$

If there exists an subseq whose $\gcd = 1$

then, $\gcd$ of an entire array $= 1$

[if, $\gcd$ of entire array $= 1$
 then, return true else return false]

TO DO $\rightarrow$ find gcd of a given array.

$$TC \Rightarrow O(N \log(\max(arr)))$$

$\Rightarrow$ Power Research
and Facebook

Given N positive distinct elements, choose $arr[i]$ & $arr[j]$,
such that $(i \neq j)$. If $abs(arr[i] - arr[j])$ is not present in
array, then insert $abs(arr[i] - arr[j])$. Keep repeating the
insertion, until no more insertions are possible, return the
min^m element in final array

ex $\Rightarrow arr \Rightarrow [6, 2, 12, 8, 4, 10] \Rightarrow min = 2$

$arr \Rightarrow [7, 9, 5, 2, 4, 3, 1, 8, 6] \Rightarrow min = 1$

$arr \Rightarrow [14, 10, 4, 6, 8, 2, 12] \Rightarrow min = 2$

$min = gcd \text{ of all element}$

Q what will be the length of the final array

minE = gcd of i/p array

maxE = max of i/p array

$$\boxed{\text{len} = \frac{\text{max}}{\text{min}} \Rightarrow \frac{\text{max(arr)}}{\text{gcd(arr)}}$$

